Siyuan (Max) Meng

Assistant Professor, Wuhan University of Science and Technology
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Academic Appointment

- **Wuhan University of Science and Technology**, Wuhan, China Nov. 2026 – Present
  Assistant Professor, School of Automotive and Transportation Engineering
  Research directions include roadside intelligent perception, transportation safety, infrastructure-centric world models, and transportation system optimization.

Education

- **University of Massachusetts Amherst**, USA Sep. 2023 – Sep. 2026
  Ph.D. in Civil Engineering (Transportation Engineering), Fully Funded GPA: 3.97/4.0
  Advisor: Dr. Chengbo Ai
  Dissertation: *Quality-Aware Multi-LiDAR Framework for Traffic Perception and Safety Analysis*
- **Chang'an University**, China Sep. 2020 – Jul. 2023
  M.S. in Transportation Engineering GPA: 3.76/4.0
  Advisor: Dr. Qiang Bai
  Thesis: *Research on Methodology for Highway Maintenance and Rehabilitation Decision-making Based on Integrated Asset Management*
- **Chang'an University**, China Sep. 2016 – Jul. 2020
  B.S. in Roadway, Bridge, and River-Crossing Engineering GPA: 3.74/4.0
  Thesis: *Comprehensive design of a newly constructed expressway in Jiangsu Province*

Research Profile

My research focuses on **roadside intelligent perception and transportation safety**, including multi-LiDAR and multi-sensor fusion, 3D object detection and tracking, quality-aware traffic conflict analysis, and uncertainty-aware perception. Building on this work, I am developing an **Infrastructure-Centric World Model (I-WM)** research direction that leverages the temporal depth of fixed roadside sensing to support richer traffic-scene representation, reasoning, and decision making. A complementary research thrust studies transportation system optimization under uncertainty, including EV routing, charging, and asset management.

Research Experience

- **Graduate Research Assistant**, UMass Amherst Sep. 2023 – Sep. 2026
  Advisor: Dr. Chengbo Ai. Doctoral research centered on two thrusts:
  - *Quality-aware roadside multi-LiDAR perception and safety:* an end-to-end pipeline spanning background modeling, unsupervised 3D detection, robust multi-point tracking, and quality-aware traffic conflict analysis, with sensor-level reliability propagated through the perception stack.
  - *Transportation system optimization under uncertainty:* robust model predictive control and system-optimal routing/charging methods for electric and mixed autonomous-human transportation systems.
- **Graduate Research Assistant**, Chang'an University Sep. 2020 – Jul. 2023
  Advisor: Dr. Qiang Bai. Developed multi-objective optimization methods for network-level pavement and integrated asset management, addressing segment grouping, traffic dynamics, environmental impacts, and uncertainty.

Publications

Journal Articles

1. **Meng, S.**, Parashar, P., Yang, Y., & Ai, C. (2027). FRGB3D: fast reliability-weighted Gaussian background modeling for roadside LiDAR traffic monitoring. *Journal of Computing in Civil Engineering*, 41(1), 04026129. DOI
2. **Meng, S.**, Zhang, Y., Raucci, D., & Ai, C. (2026). MulDet3D: multi-objective optimization based unsupervised object detection for multiple roadside LiDARs. *Journal of Transportation Engineering Part A: Systems*, 152(10), 04026100. DOI

3. **Meng, S.**, Bai, Q., & Shao, Y. (2026). A multi-objective optimization method for integrated road asset management considering traffic dynamics and environmental impacts. *Transportation Research Record*, 0(0), 03611981261441294. DOI
4. Cao, R., **Meng, S.**, Feng, S., Dong, S., & Bai, Q. (2024). Network-level pavement maintenance and rehabilitation planning considering uncertainties. *International Journal of Pavement Engineering*, 25(1), 2340001. DOI
5. **Meng, S.**, Bai, Q., Chen, L., & Hu, A. (2023). Multiobjective optimization method for pavement segment grouping in multiyear network-level planning of maintenance and rehabilitation. *Journal of Infrastructure Systems*, 29(1), 04022047. DOI
6. Bai, Q., Wu, S., Cao, R., **Meng, S.**, & Xu, Z. (2022). Review on passenger boarding process optimization at civil airports. *Journal of Traffic and Transportation Engineering* [in Chinese], 22(04), 68–88. DOI
7. Hu, A., Bai, Q., Chen, L., **Meng, S.**, Li, Q., & Xu, Z. (2022). A review on empirical methods of pavement performance modeling. *Construction and Building Materials*, 342, 127968. DOI
8. Bai, Q., Shao, Y., **Meng, S.**, Wang, Y., Chen, X., & Feng, H. (2022). Algorithm for identifying violation behavior of vehicles of picking up passengers at airport departure floor based on surveillance video. *Journal of Chang'an University (Natural Science Edition)* [in Chinese], 42(04), 73–86. DOI

Preprint

1. **Meng, S.** & Ai, C. (2026). Infrastructure-centric world models: bridging temporal depth and spatial breadth for roadside perception. *arXiv preprint arXiv:2604.17651*. Link

Conference Proceedings

1. Fang, Z. & **Meng, S.** (2025). Estimating travel mode choices of residents in Stockholm region based on multinomial logit model and mixed logit model. *Proc. SPIE 13575, STCE 2024*, 135753V. DOI
2. Li, C. & **Meng, S.** (2025). Analysis of expressway maintenance decision based on game theory. *Proc. SPIE 13575, STCE 2024*, 135751T. DOI
3. **Meng, S.**, Shao, Y., Cao, R., & Wu, S. (2022). Multi-objective optimization of network-level pavement maintenance decisions considering user costs [in Chinese]. In *Proc. World Transport Convention (WTC 2022)*, pp. 693–701. DOI

Manuscripts Under Review / Submitted

1. **Meng, S.**, Cheng, R., & Ai, C. MulTrack3D: robust multi-point tracking for multiple roadside LiDARs via voting-based consensus. *Automation in Construction*. **(Under Review after Major Revision)**
2. **Meng, S.**, Yang, Y., Christofa, E., Knodler, M., & Ai, C. Quality-aware vehicle conflict detection with spatial precision enhancement for roadside multi-LiDAR systems. *Accident Analysis & Prevention*. **(Submitted)**
3. **Meng, S.** & Ai, C. Station information and request timing for reducing fast-charging queues on intercity highway corridors. *Transportation Research Part D: Transport and Environment*. **(Submitted)**
4. **Meng, S.** & Ai, C. When does queue-aware charging-station choice pay off in mixed autonomous and human-driven electric ride-hailing fleets? *IEEE Transactions on Intelligent Transportation Systems*. **(Submitted)**
5. **Meng, S.** & Ai, C. Calibration drift and colorization error in roadside camera–LiDAR systems: a saturating law and site-specific recalibration budgets. *IEEE Transactions on Intelligent Transportation Systems*. **(Submitted)**
6. **Meng, S.** & Ai, C. Tracking quality determines traffic conflict detectability in roadside LiDAR systems: evidence from ClusterTrack3D. *Journal of Computing in Civil Engineering*. **(Submitted)**

Presentations

1. **Meng, S.** (2026). Quality-aware vehicle conflict detection for roadside multi-LiDAR systems. *Safe Mobility Conference 2026*, Seattle, WA.
2. **Meng, S.** (2026). FRGB3D: fast reliability-weighted Gaussian background modeling for roadside LiDAR traffic monitoring. *TRB 2026 Annual Meeting*, Washington, D.C.
3. **Meng, S.** (2025). MulDet3D: multi-objective optimization based unsupervised object detection for multiple roadside LiDARs. *INFORMS Annual Meeting 2025*, Atlanta, GA.
4. **Meng, S.** (2025). Multi-objective optimization method for integrated roadway asset maintenance and rehabilitation considering traffic impacts. *TRB 2025 Annual Meeting*, Washington, D.C.
5. **Meng, S.** (2024). Multiobjective optimization method for pavement segment grouping in multiyear network-level planning of maintenance and rehabilitation. *INFORMS Analytics Conference 2024*, Or-

lando, FL.

6. **Meng, S.** (2022). Multi-objective optimization of network-level pavement maintenance decisions considering user costs. *World Transport Convention 2022*, Wuhan, China. **Excellent Paper Award**

Teaching Experience

- **Teaching Assistant**, UMass Amherst Sep. 2024 – Jan. 2025
  Course: Programming for Civil Engineers (Python-based).
  - Designed homework assignments and exam problems for an undergraduate Python programming course.
  - Led hands-on lab sessions and guided students through coding and debugging exercises.
  - Graded assignments with detailed feedback and held regular office hours to support student learning.

Patents & Software

1. **Meng, S.**, Ai, C., & Xie, Y. (2026). System and method for unsupervised object detection using multiple LiDAR sensors. U.S. Provisional Patent Application No. 64/085,047, filed June 8, 2026.
2. Bai, Q., Shao, Y., **Meng, S.**, et al. (2025). Deep learning-based automatic detection of illegal vehicles at airport departure levels. Granted Chinese Invention Patent No. CN114821490B [in Chinese].
3. **Meng, S.**, et al. (2023). Airport curbside illegal vehicle monitoring system. Software Copyright [in Chinese].

Honors & Awards

1. Outstanding Graduate Student, Chang'an University (2023)
2. First Prize Scholarship, Chang'an University (2021)
3. Second Prize, English Speech Contest, Northwest China (2021)
4. Excellent Undergraduate Thesis, Chang'an University (2020)

Professional Service

Reviewer for peer-reviewed journals including *IEEE Transactions on Multimedia*, *Journal of Transportation Engineering Part A: Systems*, *Journal of King Saud University Computer and Information Sciences*, *International Journal of Pavement Engineering*, *International Journal of Pavement Research and Technology*, *Transportation Research Record*, and *Cluster Computing*.

Technical Skills

Programming: Python, C++, MATLAB, LaTeX **ML/CV:** PyTorch, TensorFlow, OpenCV, Open3D **Simulation/Robotics:** CARLA, SUMO, RoadRunner, Blender, ROS2, Unity **Other:** GIS, SLAM, optimization modeling, LiDAR point cloud processing

References

- Dr. Chengbo Ai, University of Massachusetts Amherst — chengbo.ai@umass.edu
- Dr. Qiang Bai, Chang'an University — qiangbai@chd.edu.cn
- Dr. Michael Knodler, University of Massachusetts Amherst — mknodler@umass.edu
- Dr. Eleni Christofa, University of Massachusetts Amherst — echristofa@engin.umass.edu
- Dr. Hao Zhang, University of Massachusetts Amherst — hao.zhang@umass.edu